FIREWALL BASIC THEORY

1. Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

Ans:-

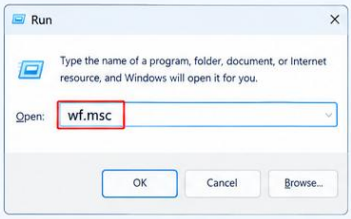
1. **Open the firewall:** Click Start, type "Windows Defender Firewall with Advanced Security" and open it.
2. **Choose Outbound Rules:** In the left pane click Outbound Rules, then in the right pane click New Rule.
3. **Rule Type:** Select Program and click Next.
4. **Program:** Select "This program path" and enter the Spotify path, for example C:\Users\<username>\AppData\Roaming\Spotify\Spotify.exe (use Browse to find it). Click Next.
5. **Action:** Select "Block the connection" and click Next.
6. **Profile:** Keep Domain, Private and Public ticked and click Next.
7. **Name:** Type "Block Spotify" and click Finish.
8. **Test:** Open Spotify. It cannot connect to the Internet any more (songs do not load). To allow it again, delete or disable the rule.

Block Spotify Outbound Traffic – Windows Defender Firewall

Create a new outbound rule to block all internet traffic from the Spotify desktop app.

1 Open Windows Defender Firewall

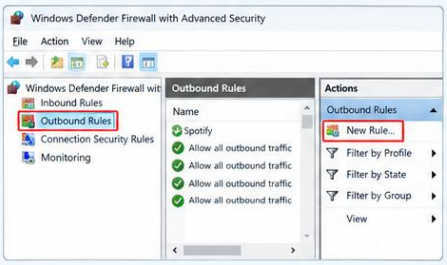
Press **Win + R**, type **wf.msc** and press Enter.



- This will open Windows Defender Firewall with Advanced Security.

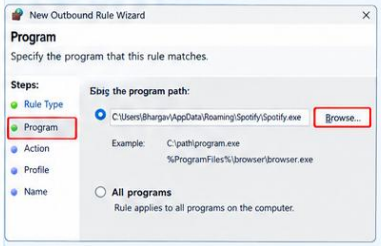
2 Go to Outbound Rules

In the left panel, select **Outbound Rules**.
Then click **New Rule...** in the right panel.



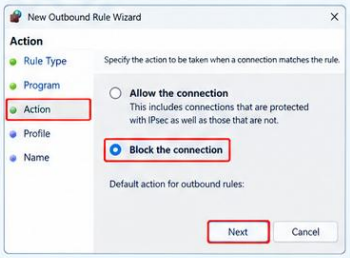
3 Select the Spotify program

1. Select **Program** and click **Next**.
2. Choose **This program path** and click **Browse**.
3. Find and select **Spotify.exe**.



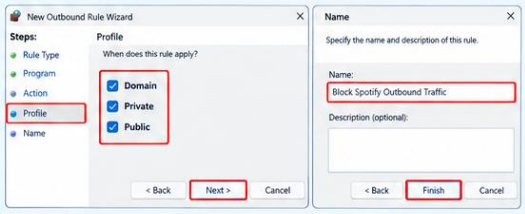
4 Block the connection

Click **Next**, select **Block the connection**, then click **Next**.



5 Apply and save

1. Select **Domain**, **Private** and **Public** profiles.
2. Click **Next**.
3. Enter the rule name: **Block Spotify Outbound Traffic**.
4. Click **Finish**.



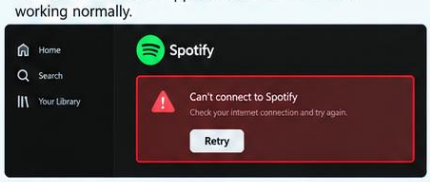
6 Verify the rule

1. Open **Outbound Rules**.
2. Find **Block Spotify Outbound Traffic**.
3. Confirm it is **Enabled** and **Action** is **Block**.



Result

Spotify's network connections matching the rule will be blocked. Other applications should continue working normally.



Note

- If Spotify was installed through the Microsoft Store, its executable path can differ.
- This rule blocks outbound traffic for the selected executable; it may not block every Spotify-related process or browser-based Spotify traffic.
- You can always disable or delete the rule later if needed (in Outbound Rules).

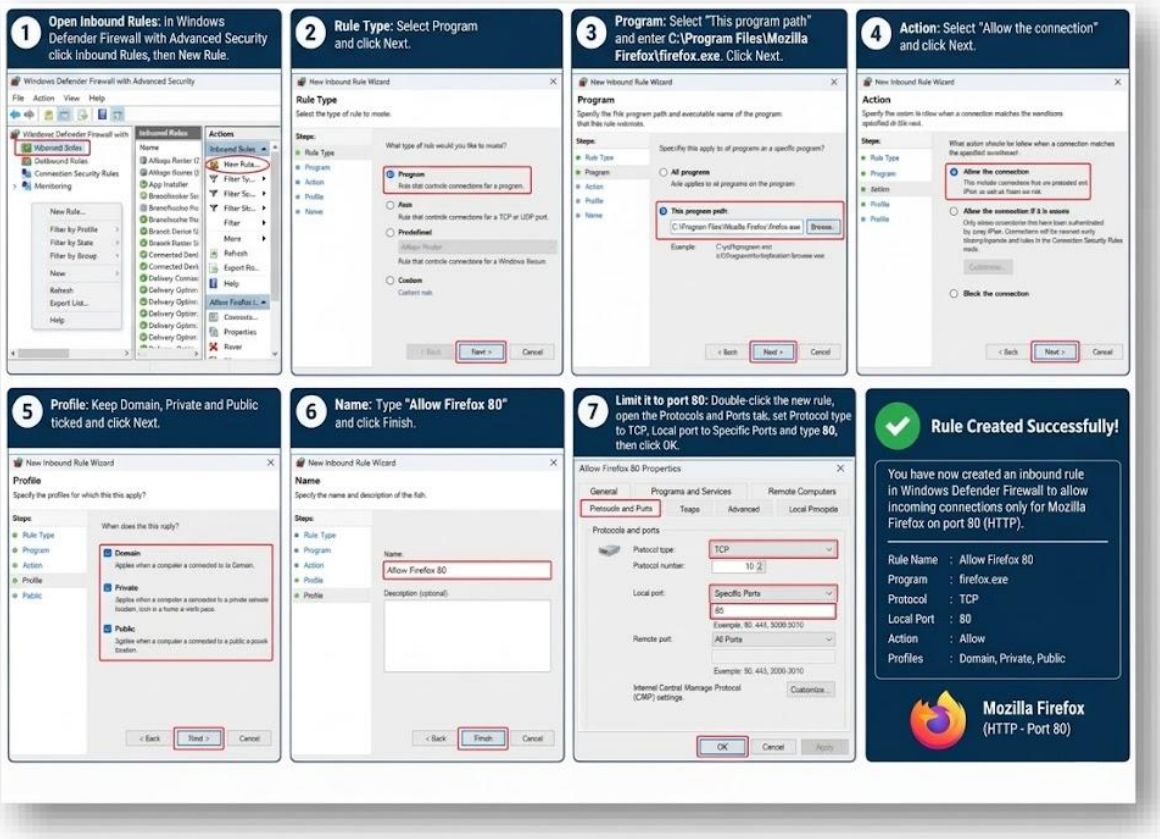
Final Rule Summary

Rule Name	Block Spotify Outbound Traffic
Direction	Outbound
Program	Spotify.exe
Action	Block
Profiles	Domain, Private, Public
Enabled	Yes

2. Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

Ans:-

1. **Open Inbound Rules:** In Windows Defender Firewall with Advanced Security click Inbound Rules, then New Rule.
2. **Rule Type:** Select Program and click Next.
3. **Program:** Select "This program path" and enter C:\Program Files\Google\Chrome\Application\chrome.exe. Click Next.
4. **Action:** Select "Allow the connection" and click Next.
5. **Profile:** Keep Domain, Private and Public ticked and click Next.
6. **Name:** Type "Allow Chrome 443" and click Finish.



3. Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall.

Hint: Use the 'Scope' tab when creating a new rule to specify the IP address.

Ans:-

Example IP address: 123.45.67.89

* Open Windows Defender Firewall: Press Win + R, type wf.msc, and press Enter.
* Create a new rule: Select Outbound Rules in the left panel, then click New Rule… in the right panel.
* Select the rule type: Choose Custom and click Next.
* Select the program: Choose All programs and click Next.
* Configure protocols and ports: Leave the default settings unchanged and click next.
* Open the Scope tab settings: Under Which remote IP addresses does this rule apply to? Select These IP addresses and click Add.
* Enter the IP address: Type 123.45.67.89, click OK, and then click next.
* Block the connection: Select Block the connection and click next.
* Select profiles: Check Domain, Private, and Public, then click next.
* Name the rule: Enter Block IP 123.45.67.89 and click Finish.
* Verify the rule: Find the rule under Outbound Rules and ensure it is enabled.

Note: To block both incoming and outgoing traffic, create a separate rule under Inbound Rules using the same IP address in the Scope settings.

4. Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

Ans:-

Point	Hardware firewall	Software firewall
Where it works	A separate device placed between the Internet and the network. It protects every computer behind it.	A program installed on one computer. It protects only that computer.
Performance	Has its own processor and memory, so it handles heavy traffic from many users without slowing computers.	Uses the computer's own CPU and RAM, so it can slightly slow the computer.
Cost and management	Costly, and needs trained staff, but is managed centrally for the whole network.	Cheap or free (Windows Defender comes with Windows), easy to set up, but must be set on each computer separately.
Use case	The gateway of a large office or college, protecting hundreds of computers.	A personal laptop at home or on public WiFi, to control which apps can connect.